

# VAGMIN VISWANATHAN

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## EDUCATION

**Dartmouth College**, Hanover, NH

**GRADUATION - May 2026**

*Bachelor of Arts in Computer Science and Physics*

**GPA 3.85/4.00**

- **Relevant Coursework:** Data Structures and Algorithms, Deep Learning, Combinatorial Optimization, Systems Programming, Cryptography, Probability, Complex Networks, Game Theory, Quantum Computing, Linear Algebra
- **Organizations:** President of Quantum Computing Club, Events Chair of West House Executive Council, ACM

## EXPERIENCE

**Amazon**, Seattle

**Jun 2025 - Sep 2025**

*Software Development Engineering Intern*

- Architected agentic onboarding system with **AWS Bedrock**, reducing onboarding time by **87%** and saving **\$400k/year**
- Built multimodal knowledge graphs and **RAG** system with Amazon OpenSearch vector store, achieving **97%** accuracy
- Developed an orchestration layer with **MCP** tool servers and a **CI/CD** pipeline to reduce manual deployments by **60%**
- Implemented **RLHF** and fine-tuned evaluation model to optimize hyperparameters, increasing accuracy by **12%**

**Millennium Management**, New York City

**Jun 2024 - Aug 2024**

*Data Science Intern*

- Engineered **ETL** pipeline to process **24k SEC** filings annually with **98%** accuracy, enabling faster investment signals
- Created a candidate profile surfacing tool with resume retrieval system for **15k+** queries via **RAG** with **83%** accuracy
- Produced Tableau dashboards for investment managers, improving due-diligence workflow efficiency by **22%**
- Analyzed competitor hiring trends using candidate employment histories generated by LLMs, sourcing **12%** more leads

**FitzLab**, Thayer School of Engineering at Dartmouth

**Aug 2023 – May 2024**

*Presidential Scholar*

- Engineered a high-performance simulation package in QuTiP to model quantum systems, cutting runtimes by **82%**
- Prototyped graph attention networks for error correction at scale, achieving results within **5%** of leading algorithms
- Optimized experimentation overhead by **30%** with robust benchmarking suite for three quantum hardware platforms

**Dartmouth Tech Consulting Group**, Hanover

**Sep 2022 – Nov 2022**

*Software Engineering Intern*

- Deployed Python-based web crawler and data pipeline to extract and normalize around **50,000** consumer profiles
- Delivered ML-driven customer segmentation model for targeted advertising, increasing revenues for client by **12%**

## PROJECTS

**Twitter Sentiment Analysis for Stock Market Prediction** | *Python*

- Engineered a sentiment analysis pipeline to scrape **1M+** tweets from Twitter, Reddit, and Truth Social with options data
- Applied **FinBERT** and transformer based models to classify investor sentiment, accounting for lead-lag effects and volatility surface dynamics using time series methods like **LSTM** and **GARCH** to study market microstructures
- Won first in applied math undergraduate poster session (out of 80 teams) for novel sentiment-driven market analysis

**Dynamical Decoupling Simulations** | *Python*

- Demonstrated agent-driven scheduling outperformed classical algorithms by **63%** in qubit coherence  $T_2^*$  times
- Developed a **RL** actor critic multi-agent framework for data-driven discovery of quantum error correction strategies
- Implemented pulse-level schedules in Qiskit over cloud based quantum hardware to characterize baseline error rates

**Secure File Transfer Protocol** | *C*

- Produced an encrypted peer-to-peer file transfer and storage CLI app, supporting upload/download of **1,000+** files
- Programmed a **TLS**-like transport layer with **<50 ms** median latency across 10 concurrent sessions without packet loss
- Integrated AES-GCM and RSA constructed from primitives for confidentiality, integrity, and origin authentication

**Tiny Search Engine** | *C*

- Coded a parallelized web crawler and indexer capable of storing **10k** pages and achieving **20 ms** query retrieval latency
- Designed a fuzzy query engine using Hypertext Induced Topic Search algorithm for ranking functionality

**Trigram Part of Speech (POS) Tagging Model** | *Java*

- Built a **Hidden Markov Model** with Viterbi decoding to perform sequence tagging on **500k** tokens with **96%** accuracy
- Architected a modular model-serving layer for POS tagging, enabling integration into downstream NLP tasks

**Congressional Twitter Polarization** | *Python*

- Leveraged data ingestion pipeline to construct interaction network of 535 politicians from around **80,000** tweets
- Simulated Bayesian updates on Erdős-Rényi models to study polarization dynamics and misinformation spread

## SKILLS

**Languages:** Python, Java, C/C++, SQL, TypeScript, Bash, Javascript, CSS, HTML

**Technologies:** PostgreSQL, React, Node.js, AWS (SageMaker, Bedrock, CloudFormation, DynamoDB, Lambda, EC2, S3)

**Frameworks:** LangGraph, TensorFlow, PyTorch, Tableau, Airflow, Pandas, Seaborn, Power BI, neo4j, Qutip, Flutter, Git